Chapter 10 Notes

Section 1

Exercise 1.1

Suppose that (A, R) is isomoprhic to (A', R') under ϕ. In fact, the biconditional does not need to be proven for any of these - proving the implication also proves the converse.

(a) Suppose (A, R) is proper. Then, for every $x \in A$, x^* is a set. Now consider any $x \in A'$. If $y \in x^*$, then $\phi^{-1}(y) \in \phi^{-1}(x)^*$. And, if $y \in \phi^{-1}(x)^*$, then $\phi(y) \in x^*$. Thus there is a 1-1 correspondence between x^* and $\phi^{-1}(x)^*$, and because $\phi^{-1}(x)^*$ is a set, so is x^* (by the axiom of substitution).

(b) Suppose (A, R) is extensional. Then, for all elements x, y in A, if $x \neq y$, then there is at least one element z in A such that z is a component of one of x or y, but not of both. Consider any elements x, y in A'. If $x \neq y$, then $\phi^{-1}(x) \neq \phi^{-1}(y)$, meaning, because (A, R) is extensional, there is some $z \in A$ such that z is a component of one of $\phi^{-1}(x)$ or $\phi^{-1}(y)$, but not of both. But, because (A, R) and (A', R') are isomorphic, it follows that $\phi(z)$ is a component of x or y, but not both, meaning (A', R') is extensional.

(c) Suppose (A, R) is well founded. Thus every subclass of A has an initial element. Consider any subclass B' of A'. Then the class B of all $x \in A$ such that $\phi(x) \in B'$ is in a 1-1 correspondence to A'. Then B has some initial element x. Then $\phi(x) \in B'$. If, for any $y \in \phi(x)^*$, $y \in B'$, we would have $\phi^{-1}(y) \in B$. But $\phi^{-1}(y) \in x^*$, contradicting the fact that x is an intial element of B. Hence $\phi(x)^* \cap B = \emptyset$, meaning (A', R') is extensional.

Section 2

Notes

Theorem 2.7 requires a bit of clarification. It is remarked that "Since every element of A has V-rank then A is well founded (Theorem 3.4 of Chapter 8), hence every element of A has A-rank, as we have seen above." This is not fully accurate, as the paragraph above shows that any relational system $(A, \in)$ where A is *transitive* and well founded is such that every element of x has A-rank. However, in Theorem 2.7, we are not assuming A is transitive.

This is however not an issue. The reason why A being transitive and well founded means every element has A-rank is because, in that case, for every subset x of A, $p(x) = \mathcal{P}(x) \cap A$ (with respect to the relational system $(A, \in)$). But, even if A is not transitive, when considering the relational system $(A, \in)$, $p(x)$ is a set for every subset x of A, as $p(x) \subseteq \mathcal{P} \cap A$.